Firm Entry and AI: Evidence from the US

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Preliminary and incomplete. Comments welcome.

Abstract

[Abstract placeholder. Summarise: motivation, data, method, main finding, contribution. Roughly 150–200 words.]

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1 Introduction

The arrival of large language models—most visibly the public release of ChatGPT in November 2022—has revived interest in how general-purpose technologies reshape labour markets and firm dynamics (Felten et al., 2021, Fossen et al., 2024). Most of this debate focuses on *workers*: which jobs are at risk, and for whom. But a parallel question may be equally consequential: does AI alter the incentives to *start* a business?

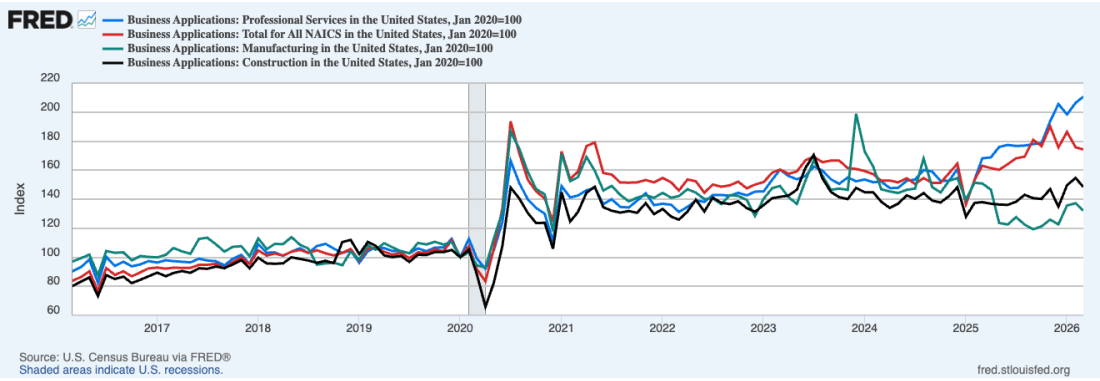

Figure 1: Business applications indexed to January 2020 = 100 for selected sectors. Source: U.S. Census Bureau via FRED (https://fred.stlouisfed.org/graph/?g=1UHD5). Shaded area indicates the COVID-19 recession (Feb–Apr 2020).

Figure 1 offers a first look. After the COVID-19 shock, aggregate business applications surged to historic highs and have continued rising through 2025. But the expansion is not uniform. Professional and Business Services—among the most AI-exposed sectors by any measure—have pulled sharply away from the aggregate trend. Manufacturing and Construction, where tasks are less amenable to language-model automation, have lagged behind. This heterogeneity motivates the present paper.

Lower entry costs, improved productivity for high-skilled tasks, and easier access to specialised knowledge may all reduce barriers to entrepreneurship in AI-exposed sectors (Brynjolfsson et al., 2018). Conversely, if AI primarily benefits large incumbents, competitive pressures could *raise* barriers to entry. Distinguishing between these channels requires variation both across sectors and over time—variation the public release of ChatGPT conveniently provides.

We use monthly sector-level business application counts from the Census Bureau Business Formation Statistics (Haltiwanger et al., 2020) together with the AI Industry Exposure (AIIE) index of Felten et al. (2021) to estimate a difference-in-differences (DiD) model of the form

$$\log(\mathrm{BA}_{it}) \;=\; \alpha_i + \gamma_t + \beta\,(\mathrm{Post}_t \times \text{AI Exposure}_i) + \varepsilon_{it}, \tag{1}$$

where Post_t is an indicator equal to one from November 2022 onwards, α_i are sector fixed effects absorbing time-invariant differences in application rates, and γ_t are month fixed effects absorbing aggregate cyclical variation. The coefficient β identifies whether sectors with higher AI exposure experienced a differential change in business formation after ChatGPT.

Related literature. Fossen et al. (2024) examine AI adoption and self-employment at the individual level and find positive effects concentrated among high-skilled workers. Felten et al. (2021) construct the AIOE/AIIE index by combining O*NET task measures with AI capability scores. Haltiwanger et al. (2020) describe the Census Bureau BFS and demonstrate its value for studying business dynamism and entry margins. Brynjolfsson et al. (2018) provide a conceptual framework for mapping task-level AI capabilities to occupational and industry outcomes.

2 Data

2.1 Business Formation Statistics

The Census Bureau BFS provides monthly, seasonally adjusted counts of new Employer Identification Number (EIN) applications at the 2-digit NAICS level, derived from IRS Form SS-4 filings (Haltiwanger et al., 2020). EIN applications are a leading indicator of actual business formation: roughly half of high-propensity applicants open within eight quarters of filing. We use the official seasonally adjusted series directly and index each sector to January 2020 = 100, yielding a panel of 20 sectors observed from January 2006 through early 2026.

We define two treatment dates: November 2022 (public launch of ChatGPT) and May 2025 (widespread availability of agentic AI tools).

Figure 2 plots the index for four selected sectors. The divergence between Professional Services and goods-producing sectors is visible from approximately 2021 and widens after November 2022, consistent with a role for AI in lowering barriers to entry specifically where tasks are more cognitively intensive.

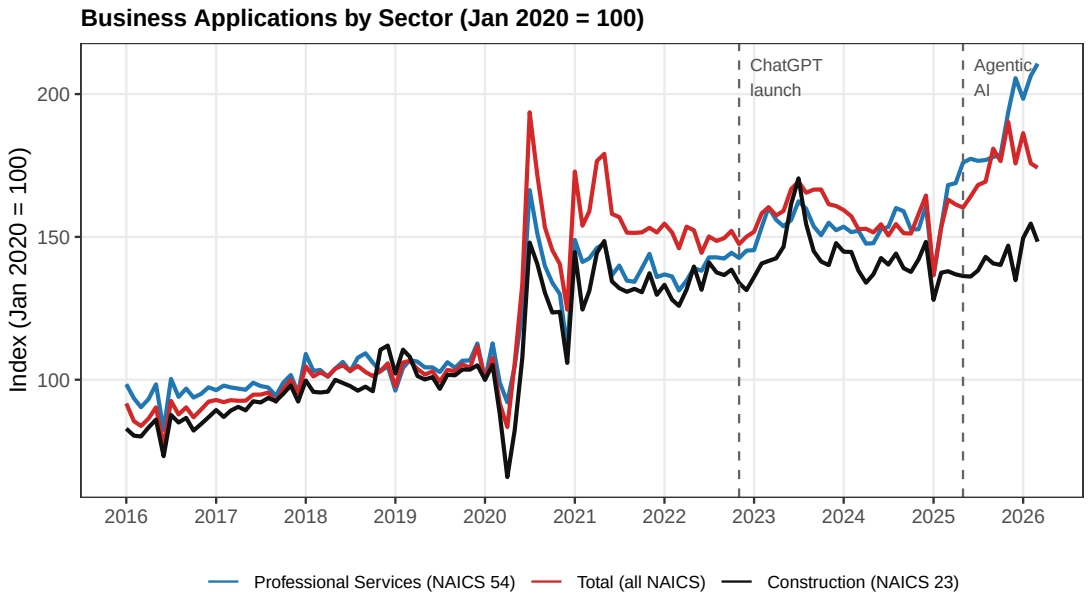

Figure 2: Business applications indexed to January 2020 = 100. Professional Services (NAICS 54), Total (all NAICS), Manufacturing (NAICS 31–33), and Construction (NAICS 23). Dashed vertical lines mark November 2022 (ChatGPT launch) and May 2025 (agentic AI tools). Seasonally adjusted. Source: Census Bureau BFS.

2.2 AI Exposure

Following Felten et al. (2021), we use the AI Industry Exposure (AIIE) index, which maps occupational AI exposure scores to 4-digit NAICS industries through BLS employment weights and aggregates to 2-digit NAICS by employment-weighted averaging. The index is constructed from 10 AI application domains benchmarked against O*NET task descriptions and captures the degree to which a sector's occupational mix is susceptible to AI automation or augmentation. Finance and Insurance (NAICS 52, AIIE = 2.05), Professional Ser-

vices (NAICS 54, AIIE = 1.59), and Information (NAICS 51, AIIE = 1.27) rank highest; Agriculture (NAICS 11, AIIE = −1.56), Transportation (NAICS 49, AIIE = −1.31), and Construction (NAICS 23, AIIE = −1.00) rank lowest. This cross-sectional variation—which is fixed before the ChatGPT shock—is the key source of identification.

3 Empirical Strategy

[Empirical strategy placeholder.]

Our baseline specification is equation (1). The identifying assumption is that, absent the ChatGPT launch, business applications would have evolved in parallel across sectors regardless of their AI exposure. We probe this assumption with an event-study specification that replaces Post_t with a full set of month-relative-to-November-2022 indicators interacted with AI Exposure_i. Parallel pre-trends would be consistent with the identification assumption.

Standard errors are two-way clustered at the 2-digit NAICS sector level using `fixest` (Bergé, 2018).

4 Results

4.1 Descriptive Evidence

Figure 3 plots mean indexed business applications by AI exposure quartile. Prior to 2020 the four quartile groups track closely together, consistent with parallel pre-trends. After the COVID-19 surge and recovery, the top quartile (high AI exposure) diverges upward, while the bottom quartile remains below the aggregate trend. The gap is most pronounced from 2024 onwards, consistent with a gradual diffusion of AI tools into entrepreneurial activity rather than an immediate discrete response.

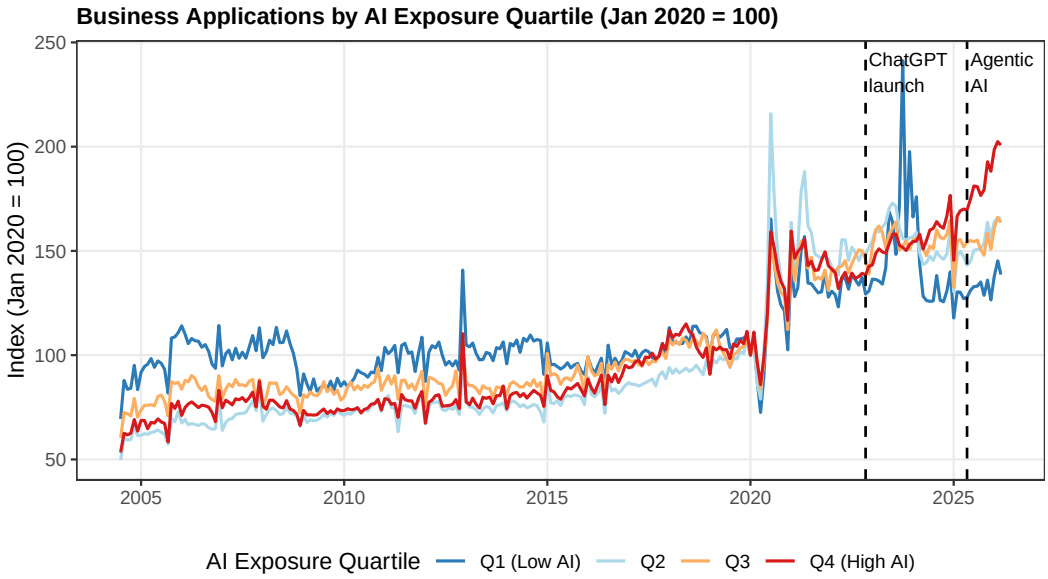

Figure 3: Business applications by AI exposure quartile (Jan 2020 = 100). Dashed vertical lines mark November 2022 (ChatGPT launch) and May 2025 (agentic AI tools).

4.2 Difference-in-Differences

Table **??** reports the DiD estimates. Columns (1)–(3) pool the entire post-November 2022 period into a single *Post* indicator; column (4) splits the post period into a ChatGPT window (November 2022–April 2025) and an agentic-AI window (May 2025 onwards). The single-period estimates are attenuated relative to the two-period specification, consistent with the event-study evidence that the effect builds gradually and becomes strongest toward the end of the sample.

4.3 Event Study

Figure 4 presents the event-study coefficients. The pre-period estimates ($t < 0$) are small and statistically indistinguishable from zero, supporting the parallel-trends assumption. Post-launch, coefficients rise gradually before the 95 % confidence interval clears zero around 2025 ($t \approx +25$ to $+30$), indicating that the differential effect of AI exposure on business formation became statistically significant roughly two to three years after the ChatGPT launch. This

Table 1: Difference-in-Differences Estimates

	(1) log(BA)	(2) BA Index	(3) BA Level	(4) Two-period
$\text{Post}_{\text{ChatGPT}} \times$ AI Exposure	0.269	35.222	416.305	0.180
	(0.248)	(33.841)	(7073.848)	(0.238)
Observations	4176	4176	4176	4176
R^2	0.984	0.673	0.901	0.984
Sector FE	X	X	X	X
Month FE	X	X	X	X

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by 2-digit NAICS sector. $\text{Post}_{\text{ChatGPT}}$: Nov 2022 – Apr 2025. $\text{Post}_{\text{Agentic}}$: May 2025 onwards.

pattern is consistent with a diffusion mechanism: entrepreneurs require time to learn the tools, identify viable business models, and translate that into actual EIN filings. A visible second inflection around $t = +30$ (May 2025, the emergence of widely available agentic AI tools) suggests a possible amplification of the initial shock, which the two-period DiD in column (4) of Table **??** is designed to capture.

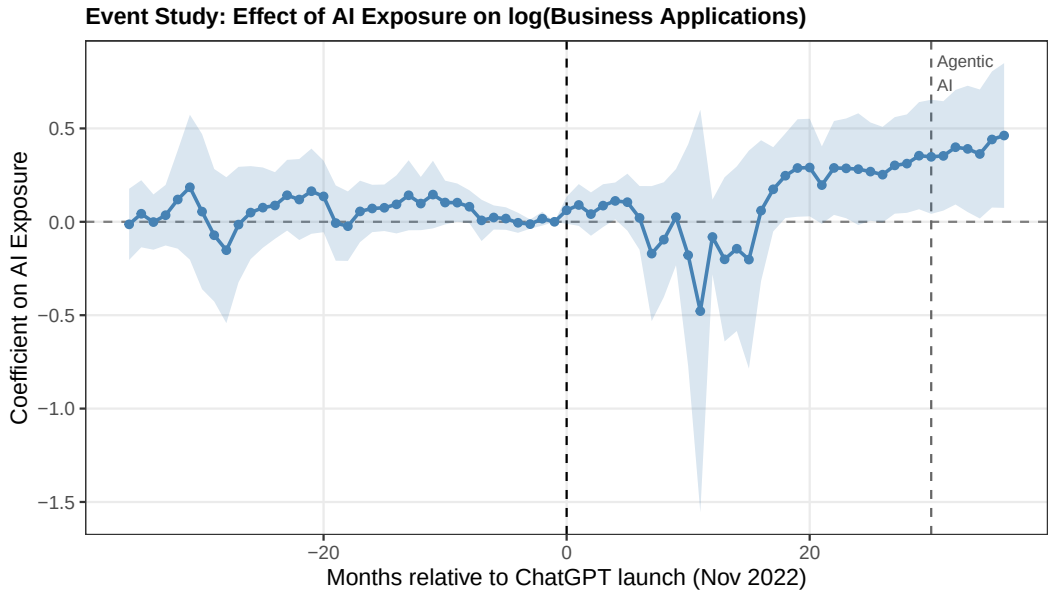

Figure 4: Event-study estimates: effect of AI exposure on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on sector-clustered standard errors. The second dashed line marks May 2025 ($t = +30$, agentic AI).

5 Conclusion

[Conclusion placeholder.]

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